

Problem Set 4: Answers

Probability and Statistical Inference

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1. A government is deciding whether to introduce an employment-coaching program. Let β denote the program's effect on the probability of employment, measured in percentage points. A positive value of β means that the program raises the employment probability. The government will roll out the program only if there is evidence that it improves employment.
 - (a) Write down suitable null and alternative hypotheses for this pre-specified question about whether the program improves employment.
 - (b) Describe a Type I error and a Type II error in terms of the program and the rollout decision.
 - (c) Suppose rolling out an ineffective program would be especially costly. Holding the design and sample size fixed, explain why the government might choose a smaller significance level, and state the resulting trade-off.
 - (d) The study produces a one-sided p -value of 0.08. What decision is made at the 5% level? Explain what the value 0.08 does and does not mean.
 - (e) In a separate scenario, an analyst sees a negative estimate and proposes changing the pre-specified right-sided test to a left-sided test. Explain why this would not be a valid way to conduct the pre-specified test. How could concern about effects in either direction have been handled?

Solution.

- (a) A suitable point-null formulation is

$$H_0 : \beta = 0 \quad \text{against} \quad H_1 : \beta > 0.$$

If non-positive effects are to be treated as a single null region, one can instead write $H_0 : \beta \leq 0$ against $H_1 : \beta > 0$; the boundary value 0 determines the usual rejection cutoff.

- (b) A Type I error is concluding that the program has a positive effect, and therefore rolling it out, when the null is true. A Type II error at a particular $\beta > 0$ is failing to find sufficient evidence for rollout even though that positive effect is real.
- (c) A smaller α makes rejection harder, reducing the probability of a costly false rollout under the null. For a fixed design and a fixed positive alternative it also increases the Type II error probability and reduces power.
- (d) Since $0.08 > 0.05$, we do not reject H_0 at the 5% level. Under the null, the p -value is the probability of obtaining a test statistic at least as favorable to H_1 as the observed statistic. It is neither an 8% posterior probability that H_0 is true nor proof that the program has no effect.
- (e) Changing the direction because of the observed sign lets the data choose both the hypothesis and the evidence used to test it. The stated one-sided Type I error rate would therefore no longer apply. If important effects in either direction were possible, a two-sided test should have been specified before examining the estimate. Alternatively, a valid adjustment could be made for multiple tests that were planned in advance.

2. An estimated regression coefficient is $\hat{\beta} = 0.18$ with estimated standard error 0.07. Consider $H_0 : \beta = 0.05$. There are 76 degrees of freedom, and you may use

$$F_{76}(1.857) = 0.9663, \quad t_{0.95,76} = 1.665, \quad t_{0.975,76} = 1.992,$$

where F_{76} is the cumulative distribution function (cdf) of a t_{76} random variable.

- Calculate the test statistic.
- Test against $H_1 : \beta > 0.05$ at the 5% level in three equivalent ways: (i) using a critical value, (ii) using a p -value, and (iii) **non-examinable extension:** using the corresponding one-sided confidence set. Verify that the methods give the same decision.
- For $H_1 : \beta \neq 0.05$, carry out a two-sided 5% test using a critical value, a p -value and a 95% confidence interval. Verify that all three decisions agree.
- Explain why the two alternatives can produce different decisions from the same estimate. Under which regression assumptions and standard-error calculation is the t_{76} reference distribution exact?

Solution.

- The statistic is

$$T = \frac{0.18 - 0.05}{0.07} = 1.857 \quad (\text{to three decimal places}).$$

- For the right-sided test:

- The critical-value rule rejects when $T > t_{0.95,76} = 1.665$. Since $1.857 > 1.665$, it rejects.
- The one-sided p -value is

$$1 - F_{76}(1.857) = 1 - 0.9663 = 0.0337 < 0.05,$$

so it rejects.

- Non-examinable extension.** The corresponding 95% one-sided confidence set is

$$[\hat{\beta} - t_{0.95,76}\hat{se}, \infty) = [0.18 - 1.665(0.07), \infty) = [0.06345, \infty).$$

The null value 0.05 is outside this set, so it rejects. The three decisions agree.

- For the two-sided test, $|T| = 1.857 < 1.992$, so the critical-value rule does not reject. The two-sided p -value is

$$2[1 - F_{76}(1.857)] = 2(0.0337) = 0.0674 > 0.05.$$

The 95% confidence interval is

$$0.18 \pm 1.992(0.07) = 0.18 \pm 0.13944 = [0.04056, 0.31944].$$

It contains 0.05, so again the three methods agree on non-rejection.

- The one-sided test places all of its rejection probability in the pre-specified right tail, whereas the two-sided test divides it between two tails. The right-tail cutoff is therefore smaller. With a linear model and full column rank, the t_{n-k} reference distribution is exact conditional on X when $\varepsilon \mid X \sim N(0, \sigma^2 I_n)$ and the usual homoscedastic variance estimator is used. Here $n - k = 76$. These are the Gauss–Markov assumptions plus normal errors. With suitable large-sample conditions and the appropriate standard error, inference can instead use a normal approximation.

3. **Non-examinable extension: inverting tests.** To invert a test means to find all null values that the test would not reject. Let $c > 0$ be the appropriate critical value, and treat $\hat{se}(\hat{\beta})$ as fixed during this calculation.

- (a) Starting from the non-rejection condition

$$\left| \frac{\hat{\beta} - \beta_0}{\widehat{se}(\hat{\beta})} \right| \leq c,$$

solve for all null values β_0 that would not be rejected. Hence derive the usual two-sided confidence interval.

- (b) For tests against $H_1 : \beta > \beta_0$, non-rejection occurs when $(\hat{\beta} - \beta_0)/\widehat{se}(\hat{\beta}) \leq c$. Invert this inequality to derive the corresponding one-sided confidence set.
- (c) Use your answers to explain the statement: “a confidence set is the collection of null values not rejected by the corresponding tests.”

Solution. Write $s = \widehat{se}(\hat{\beta}) > 0$.

- (a) The two-sided non-rejection condition is equivalent to

$$-c \leq \frac{\hat{\beta} - \beta_0}{s} \leq c.$$

Multiplying by s and rearranging both inequalities gives

$$\hat{\beta} - cs \leq \beta_0 \leq \hat{\beta} + cs.$$

Thus the collection of non-rejected null values is $[\hat{\beta} - cs, \hat{\beta} + cs]$.

- (b) For the right-sided family,

$$\frac{\hat{\beta} - \beta_0}{s} \leq c \iff \beta_0 \geq \hat{\beta} - cs.$$

The non-rejected values therefore form $[\hat{\beta} - cs, \infty)$.

- (c) In each case, imagine testing every possible point null $H_0 : \beta = \beta_0$ by the same level- α rule. The confidence set keeps exactly the values of β_0 that satisfy the non-rejection inequality. This algebra is why test and confidence-set decisions cannot disagree when their reference distribution, sidedness, and significance level match.
4. Let $Y_1, \dots, Y_n$ be independent $N(\mu, 12^2)$ observations. We test $H_0 : \mu = 100$. Suppose first that $n = 36$, so the standard error of $\bar{Y}$ is 2, and consider the alternative value $\mu = 104$. Power at $\mu = 104$ means the probability of rejecting the null when the true mean is 104. Use

$$z_{0.95} = 1.645, \quad z_{0.975} = 1.960, \quad z_{0.80} = 0.842,$$

as well as $\Phi(0.355) = 0.6387$, $\Phi(0.04) = 0.5160$, and $\Phi(-3.96) \simeq 0.00004$, and $\Phi(-3.645) \simeq 0.00013$.

- (a) Derive the rejection rule and power at $\mu = 104$ for a right-sided 5% test.
- (b) Derive the rejection rule and power at $\mu = 104$ for a two-sided 5% test.
- (c) Explain the source of the difference between the two powers. What is the approximate power of the right-sided test if the true mean is instead 96?
- (d) Find the smallest integer n giving at least 80% power at $\mu = 104$ for the right-sided test.
- (e) Using the usual minimum-detectable-effect (MDE) approximation, find the required n for 80% power at $\mu = 104$ with a two-sided test. What happens to the standard error and the MDE if n is quadrupled?

Solution.

- (a) The right-sided rule rejects when

$$\frac{\bar{Y} - 100}{2} > 1.645, \quad \text{or equivalently when } \bar{Y} > 103.29.$$

Under $\mu = 104$, $(\bar{Y} - 104)/2 \sim N(0, 1)$, so

$$\begin{aligned} \text{Power}_{104}(104) &= \Pr_{104}(\bar{Y} > 103.29) \\ &= \Pr\left(Z > \frac{103.29 - 104}{2}\right) = \Pr(Z > -0.355) = \Phi(0.355) = 0.6387. \end{aligned}$$

- (b) The two-sided rule rejects outside

$$100 \pm 1.96(2) = [96.08, 103.92].$$

At $\mu = 104$ its power is

$$\begin{aligned} \Pr_{104}(\bar{Y} < 96.08) + \Pr_{104}(\bar{Y} > 103.92) &= \Phi(-3.96) + 1 - \Phi(-0.04) \\ &= 0.00004 + \Phi(0.04) \simeq 0.5160. \end{aligned}$$

- (c) Both tests have size 0.05, but the one-sided test puts all of that rejection probability in the direction $\mu > 100$. It is therefore more powerful at 104. If $\mu = 96$, right-sided rejection requires

$$Z > \frac{103.29 - 96}{2} = 3.645,$$

whose probability is about 0.00013. This very low power in the opposite direction is the price of the directional test.

- (d) With general n , the right-sided power at 104 is

$$\Phi\left(\frac{(104 - 100)\sqrt{n}}{12} - 1.645\right).$$

Requiring this to be at least 0.8 gives

$$\frac{4\sqrt{n}}{12} - 1.645 \geq 0.842, \quad n \geq \left[\frac{12(1.645 + 0.842)}{4}\right]^2 = 55.67 \dots$$

The smallest integer is $n = 56$.

- (e) The usual two-sided design approximation gives

$$n \geq \left[\frac{12(1.960 + 0.842)}{4}\right]^2 = 70.66 \dots,$$

so $n = 71$. (The formula neglects the negligible rejection probability in the tail opposite the specified alternative.) Since a standard error is proportional to $n^{-1/2}$, quadrupling n halves both the standard error and the minimum detectable effect.

5. A research team conducts $m = 20$ independent tests of different outcomes and reports the outcome that gives the strongest apparent evidence against its null hypothesis (the smallest p -value). Assume all 20 null hypotheses are true. For the exact probability calculations, assume each test's actual rejection probability equals its chosen significance level: 0.05 initially and 0.05/20 after the Bonferroni correction.

- (a) Find the probability of at least one false rejection, which is the family-wise error rate. Distinguish this probability from the expected number of false rejections.

- (b) Under a Bonferroni correction designed to control the family-wise error rate at 0.05, what level is used for each test?
- (c) Use the union bound to show that Bonferroni controls the family-wise error rate even without independence. Under independence, calculate its exact family-wise error rate.
- (d) The smallest unadjusted p -value is 0.018. Is it significant under the Bonferroni rule? Explain how pre-specifying a primary outcome changes the multiple-testing problem.
- (e) In a separate Monte Carlo exercise (a repeated computer simulation) with $R = 100,000$ repetitions, a nominal 5% right-sided test rejects 5,041 times under the null and 63,870 times at a specified alternative. Estimate size and power. Explain the different effects of increasing the number of simulation repetitions R and increasing the study sample size n .

Solution.

- (a) The probability of no false rejection is 0.95^{20} , so

$$\text{FWER} = 1 - 0.95^{20} = 0.6415 \quad (\text{approximately}).$$

The expected number of false rejections is instead the sum of the expected values of the 20 rejection indicators: $20(0.05) = 1$. One is an “at least one” probability; the other is an average count.

- (b) Bonferroni tests each outcome at level

$$\frac{0.05}{20} = 0.0025.$$

- (c) Let A_j be the event of falsely rejecting test j . Whether or not the events are independent,

$$\Pr\left(\bigcup_{j=1}^{20} A_j\right) \leq \sum_{j=1}^{20} \Pr(A_j) \leq 20(0.0025) = 0.05.$$

If the tests are independent and each has actual rejection probability 0.0025 under its true null, then

$$\text{FWER} = 1 - (1 - 0.0025)^{20} = 1 - 0.9975^{20} \simeq 0.04883.$$

- (d) Since $0.018 > 0.0025$, the result is not family-wise significant by this rule. If one outcome is genuinely designated as primary before seeing the data, its single pre-specified test need not pay a penalty for searching across all 20 outcomes. Secondary outcomes should still be described and handled transparently.
- (e) The simulated size is $5,041/100,000 = 0.05041$ and the simulated power is $63,870/100,000 = 0.63870$. Increasing R gives these Monte Carlo estimates smaller variance but does not alter the sampling distribution or power of the real study. Increasing n changes the study’s standard error and generally changes its power.